

HCAHPS Patient Analysis Project

Power BI Data Modeling

Prepared By

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Tools Used

Power BI

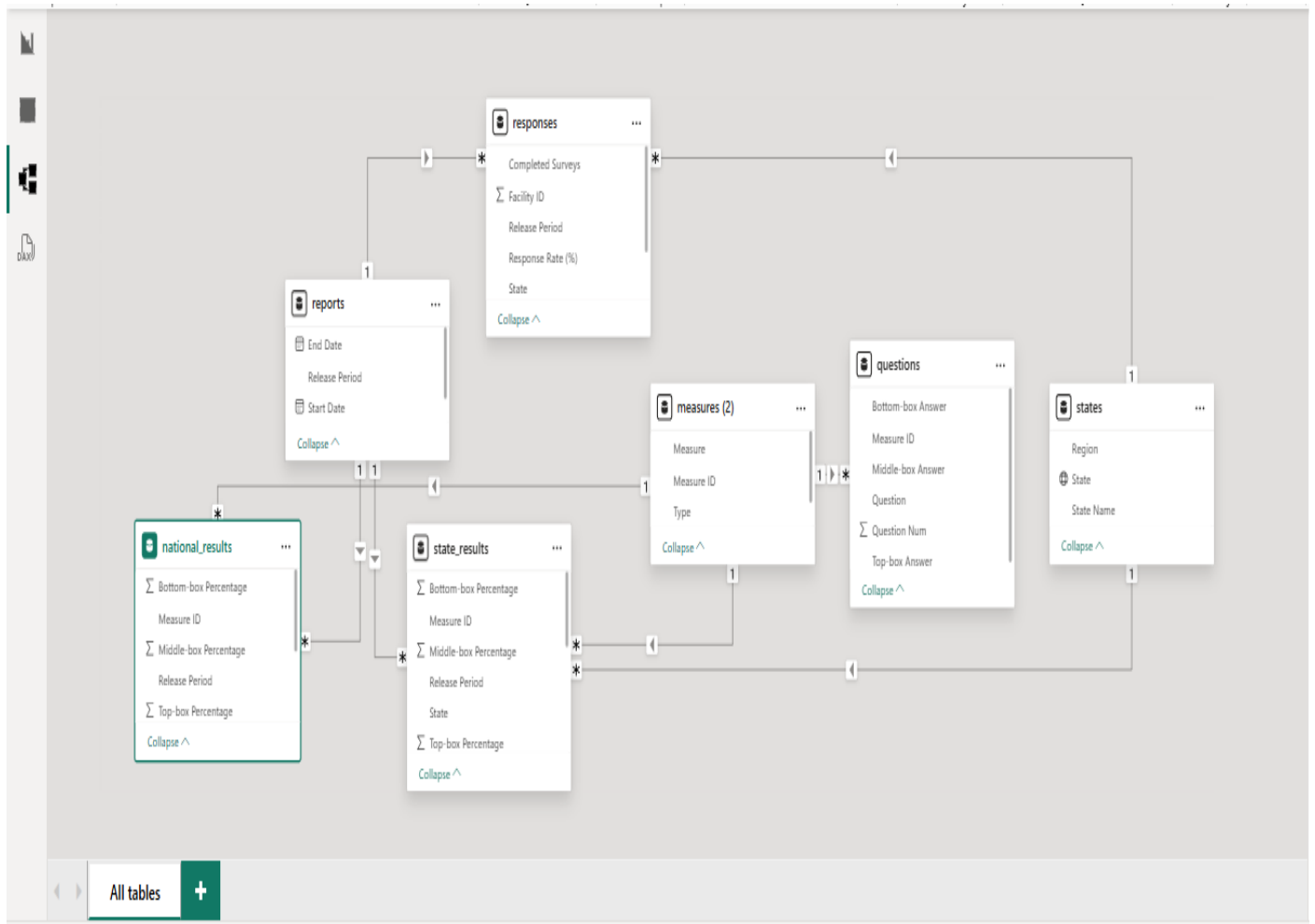
Power Query

Excel

Submission Date

6 August 2026

1. Star Schema



The Power BI data model was designed using a **Star Schema**. The model consists of fact tables containing patient satisfaction data and dimension tables containing descriptive information such as states, measures, questions, and reporting periods. This structure improves performance, simplifies relationships, and supports efficient dashboard analysis.

2. Fact Tables

Fact Table	Description
National Results	Stores national-level Top-box, Middle-box, and Bottom-box percentages for each measure and reporting period.
State Results	Stores state-level patient satisfaction percentages for each measure and reporting period.
Responses	Stores completed surveys and response rate information by state and reporting period.

3. Dimension Tables

Dimension Table	Description
States	Contains State name, State code, and Region information used for geographical analysis.
Measures	Contains Measure ID, Measure Name, and Measure Type used for patient experience analysis.
Questions	Contains survey question details associated with each Measure ID.
Reports	Contains Release Period, Start Date, and End Date used for time-based analysis.

4. Relationships

From Table	To Table	Key Used
Reports	National Results	Release Period
Reports	State Results	Release Period
Reports	Responses	Release Period
Measures	National Results	Measure ID
Measures	State Results	Measure ID
Measures	Questions	Measure ID
States	State Results	State
States	Responses	State

5. Cardinality

Relationship	Cardinality
Reports → National Results	One-to-Many
Reports → State Results	One-to-Many
Reports → Responses	One-to-Many
Measures → National Results	One-to-Many
Measures → State Results	One-to-Many
Measures → Questions	One-to-Many
States → State Results	One-to-Many
States → Responses	One-to-Many

6. Cross-filter Direction

All relationships use **Single** cross-filter direction. This allows dimension tables to filter the related fact tables while maintaining good performance and preventing ambiguous relationships.

7. Modeling Decisions

Modeling Decision	Reason
Star Schema was used	Improves performance and simplifies reporting.
Separate Fact and Dimension tables	Reduces redundancy and improves maintainability.
Reports table used as the time dimension	Enables consistent filtering across all dashboards.
States table used as the geographic dimension	Supports state and regional analysis.
Measures table used as the common dimension	Ensures consistent measure mapping across fact tables.
Questions linked through Measure ID	Associates survey questions with the relevant measures.
One-to-Many relationships	Dimension tables contain unique records that relate to multiple fact records.
Single cross-filter direction	Prevents ambiguous filter paths and improves model performance.